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R. Rahul
192521362

CSA0311 DATA STRUCTURES

ASSESSMENT TOOL - I

ANALYTICAL PROBLEM

1. Convert the following infix expression into postfix notation using a stack. Show the contents of the stack after processing each symbol.

(A + B) x (C - D) / E

2/7/7

Infix expression,

CHARACTER	STACK	OUTPUT
(	(	Empty
A	(	A
+	(+	A
B	(+	AB
)	(+)	AB+
*	*	AB+
(	*(	AB+
C	*(	AB+C
-	*(-	AB+C
D	*(-	AB+CD
)	*(-)	AB+CD-
/	/	AB+CD-/
E	/	AB+CD-/E

Empty

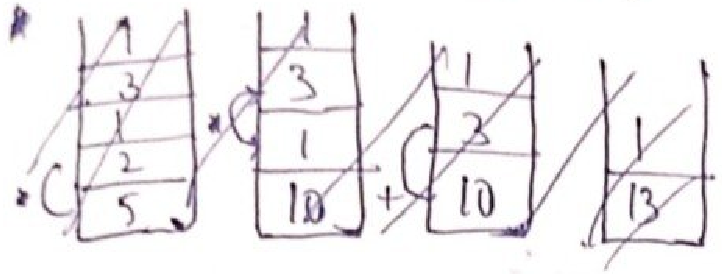
AB+CD-*E/

i) Find Postfix Expression.

AB+CD-*E/

ii) Postfix evaluation

A=5, B=2, C=1, D=3, E=1



i) 5+2=7

ii) 1-3=-2

iii) 7(-2) = -14

↓

-14 ÷ 1 = -14

2. Convert the following infix expression into postfix form. Explain how operator precedence and associativity influence the conversion.

A+B*C-(D/E+F)*G

Infix expression,

A+B*C-(D/E+F)*G

CHARACTERS	STACK	POSTFIX
A		A
+	+	A
B	+	AB
*	+*	AB
C	+*	ABC
-	-	ABC*+

		ABC*+
(	-(	ABC*+D
D	-(	ABC*+D
/	-(/	ABC*+DE
E	-(/	ABC*+DE/
+	-(+	ABC*+DE/F
F	-(+	ABC*+DE/F+
)	-(+)	
G	*	ABC*+DE/F+
*	-*	ABC*+DE/F+G
G	-*	ABC*+DE/F+G*-
Empty		

Final Postfix Expression,

ABC*+DE/F+G*-

Operator Precedence,

i) Parentheses () → Highest priority

ii) Multiplication * and Division /

iii) Addition + and Subtraction -

- +, -, *, / are left to right associative
- When two operators have the same precedence, the left operator is evaluated first.